CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

May 2012

CL 210 Hydrology and Hydraulics

Date: 17.05.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Enlist various methods for measurement of evapotranspiration. **[2]**

(b) A storm with a 10.0 cm precipitation produced a direct runoff of 5.8 cm. Given the time distribution of the storm as below, estimate the Φ-index of the storm and show it on graph paper. Also check the total rainfall excess with given value. **[5]**

Time from start (hr)	0	1	2	3	4	5	6	7	8
Cumulative rainfall (cm)	0	0.4	1.3	2.8	5.1	6.9	8.5	9.5	10

Q - 2 (a) For a basin, the area measured for each polygon, rainfall observed in that area and area enclosed between adjacent isohyets are given below. Compute the average rainfall by (i) Arithmetic mean method (ii) Thiessen polygon method and (iii) Isohyetal method. **[5]**

Area of Each Polygon (A), mm^2	557	421	233	125	450
Observed Precipitation (P), mm	122	140	150	143	129

Isohyet, mm	35	70	105	140	175
Area enclosed between adjacent isohyets (A), mm^2	220	430	150	510	476

(b) Flood-frequency computations for a river Mahi using Gumbel's method, yielded the following results: **[5]**

Return Period T (years)	**Peak flood (m^3/s)**
50	45,000
100	50,000

Estimate the flood magnitude in this river with a return period of 500 years, if s_n and $\bar{y}_n$ are constants.

(c) (i) Draw a neat sketch of weighing type rain gauge. **[2]**

(ii) Enlist the factors affecting runoff? **[2]**

OR

Q - 2 (a) For a catchment, the mean monthly rainfall and temperatures are given below. Calculate the annual runoff and annual runoff coefficient by Khosla's formula. **[5]**

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Temp (°C)	-3	4.5	30	35	45	32	29	27	22	20	10	11
Rainfall (cm)	1	2	0	0	5	10	35	27	18	3	1	2

Losses 'Lm' may provisionally assumed as,

T (°C)	4.5	-1	-6.5
Lm (cm)	2.17	1.78	1.52

(b) Given below are the ordinate of a 6-hr unit hydrograph for a catchment. Calculate the ordinates of the DRH due to rainfall excess of 1.5 cm occurring in 6-hr. Draw UH and also show the values of DRH in same graph paper. [5]

Time (hr)	0	6	12	18	24	30	36
UH ordinate (m^3/s)	0	50	100	250	370	320	220
Time (hr)	42	48	54	60	66	72	
UH ordinate (m^3/s)	120	72	50	32	16	0	

(c) Define flood and flood routing. Briefly discuss the flood forecasting techniques. [4]

Q - 3 (a) The time-drawdown data from an observation well 10.5 m from a pumped well is given in table below. The test well is pumped at the rate of 1100 lpm. Determine the constants 'T' and 'S' by Chow's method. (Refer Fig. 1) [9]

Time 't' (min)	1	2	4	8	20	40	60	80	100
Drawdown 's' (m)	0.16	0.22	0.29	0.35	0.42	0.49	0.52	0.55	0.56

(b) Explain following surface ground water investigation techniques: [5]
(i) Electrical resistivity (ii) Seismic refraction

OR

Q - 3 (a) Explain well losses and well development methods in detail. [8]

(b) Write short notes on following sub-surface ground water investigation techniques: [6]
(i) Resistivity logging (ii) Radioactive logging

SECTION – II

Q - 4 (a) Explain water conservation. Discuss in detail the techniques to conserve the surface runoff. [5]

(b) What is ground water pollution? Enlist factors affecting the same. [2]

Q - 5 (a) Differentiate between natural and artificial ground water recharge. What are the advantages of artificial ground water recharge? [6]

(b) Discuss the sources and nature of pollution from sewage and nuclear wastes. [4]

(c) State Darcy's Law. Explain the concept of conjunctive use. [4]

OR

Q - 5 (a) Discuss bacteriological and physical parameters affected to quality of ground water. [6]

(b) Discuss the sources and nature of pollution from industrial and agricultural wastes. [4]

(c) (i) Define transmissibility. (ii) An aquifer has 30 m thickness, 30% porosity and bulk modulus of compression as 10^7 N/m^2. Estimate the storage coefficient of the aquifer. Take $K_w = 2.1$ GN/m^2. [4]

Q - 6 (a) What is salt water intrusion? Explain the concept of slope and shape of interface with neat sketch. [6]

(b) It was observed in a field test that 2 hr 30 min. was required for a tracer to travel from one well to another 15 m apart. The difference in water surface elevations was 0.45 m. Samples of the aquifer between the wells indicated a porosity of 20%. Determine the permeability of the aquifer, seepage velocity and the Reynolds number for the flow, assuming, an average grain size of 1 mm and v_{water} at 27°C = 0.0075 Stoke. [5]

(c) Explain constant head permeability measurement method. [3]

OR

Q - 6 (a) (i) Which prevention measures should be taken to control the salt water intrusion? [4]

(ii) A well screen 1 m in length is located 18 m below the ground water table in an unconfined aquifer having a permeability of 25 m/day. The fresh water-sea water interface exists at a depth of 30 m below the water table. What is the maximum discharge that can be sustained from well without causing salt water to intrude into the well? [2]

(b) The average thickness of the aquifer of the confined aquifer is 25 m. It is extended over an area of 630 km^2. The piezometric surface fluctuates annually from 20 to 10 m above the top of aquifer. What ground water can be expected annually, if a storage coefficient is 0.00063? Assuming, an average well yield of 20 m^3/hr and about 180 days of pumping in a year, how many wells can be drilled in the area? [5]

(c) Explain the falling head permeability measurement method. [3]

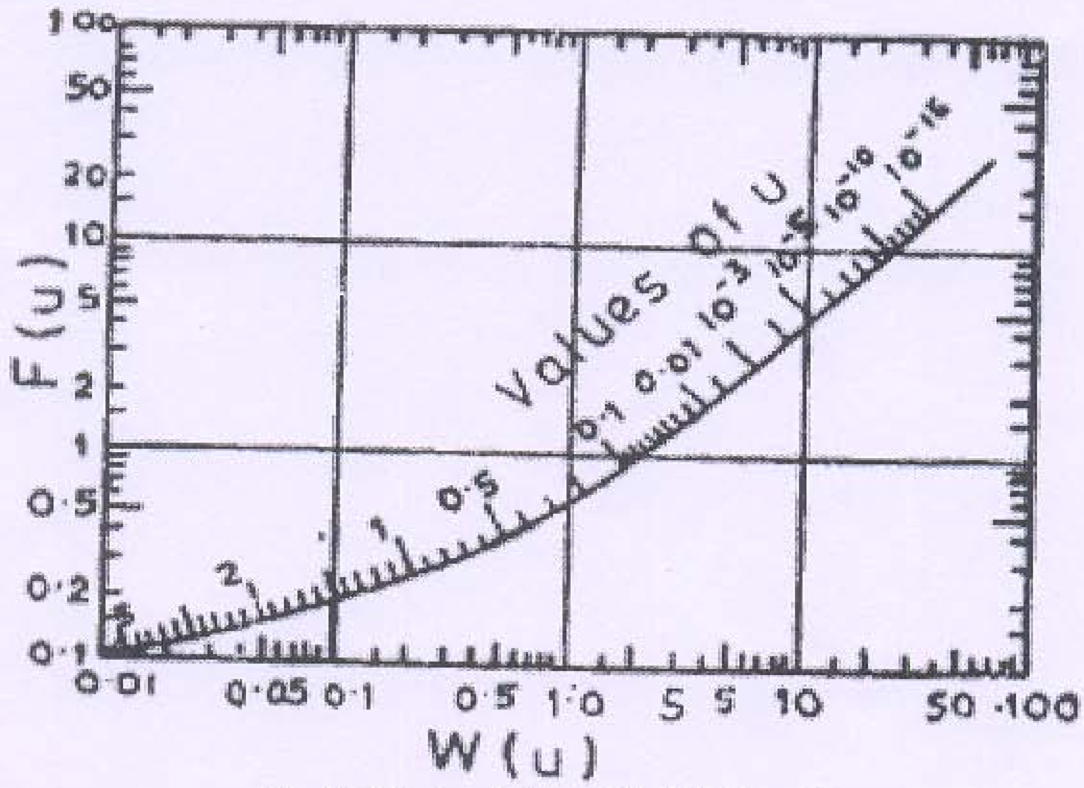

Fig. 1: Relation between F(u), W(u) and u

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